We learn about shard sizing in chapter 14. Shards and replicas make nodes, and nodes form clusters. Let's learn about these in the next section.

3.2.5 Nodes and clusters

When you launch Elasticsearch, it boots up a single instance called a *node*. Each node hosts a set of shards and replicas (the instances of Apache Lucene). The index, a logical collection to hold our data, is created across these shards and replicas. Figure 3.14 shows a single node forming a cluster.

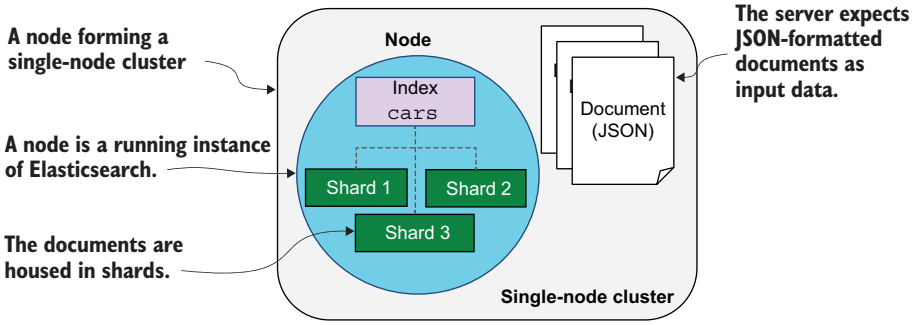


Figure 3.14 A single-node Elasticsearch cluster

While you can start multiple nodes on the same machine—thus creating a multi-node cluster—doing so is not advisable. Replicas will never be created on the same machine where their respective shards exist, leading to a situation with no backup of our data. This leaves the cluster in an unhealthy YELLOW state. To start another node on the same machine, all we need to do is make sure the data and log paths are different (check out the sidebar "Additional node on a personal machine" for details).

SINGLE-NODE CLUSTERS

When you boot up a node for the first time, Elasticsearch forms a new cluster, commonly called a *single-node cluster*. While we can use a single-node cluster for development purposes, it is far from a production-grade setup. A typical production environment has a farm of data nodes forming clusters (single or multiple) based on the data and application search requirements. If we start another node in the same network, the newly instantiated node joins the existing cluster, provided the `cluster.name` property points to the single-node cluster. We discuss production-grade setup in chapter 14.

> **NOTE** A property file named elasticsearch.yml (in the <*INSTALL_DIR*>/config directory) mentions a property called `cluster.name`, which dictates the name of our cluster. This is an important property as all nodes with the same name join together and form a cluster. For instance, if we expect a 100-node